

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a data-centric system designed to orchestrate the lifecycle of football matches, managing the relational mapping between teams, leagues, and users. It implements a strict validation layer for match states and scoring, utilizing a many-to-many relationship for user match-following. The system focuses on data integrity and query performance via targeted indexing on high-traffic filter fields.

1.2 Maturity Assessment

The project is in a state of **REFINEMENT**. While the data model and validation rules are exhaustively defined, the specifications suffer from high-severity structural gaps, specifically the complete absence of business goals, objectives, and scope definitions. Furthermore, a technical uncertainty regarding the scope of team name uniqueness remains unresolved.

1.3 Technical Stack

- **Primary Keys:** UUID (128-bit universally unique identifiers)
- **Temporal Standard:** UTC (Coordinated Universal Time)
- **Data Types:** DateTime, Enum, Boolean, Integer, String, URL

1.4 Architectural Constraints

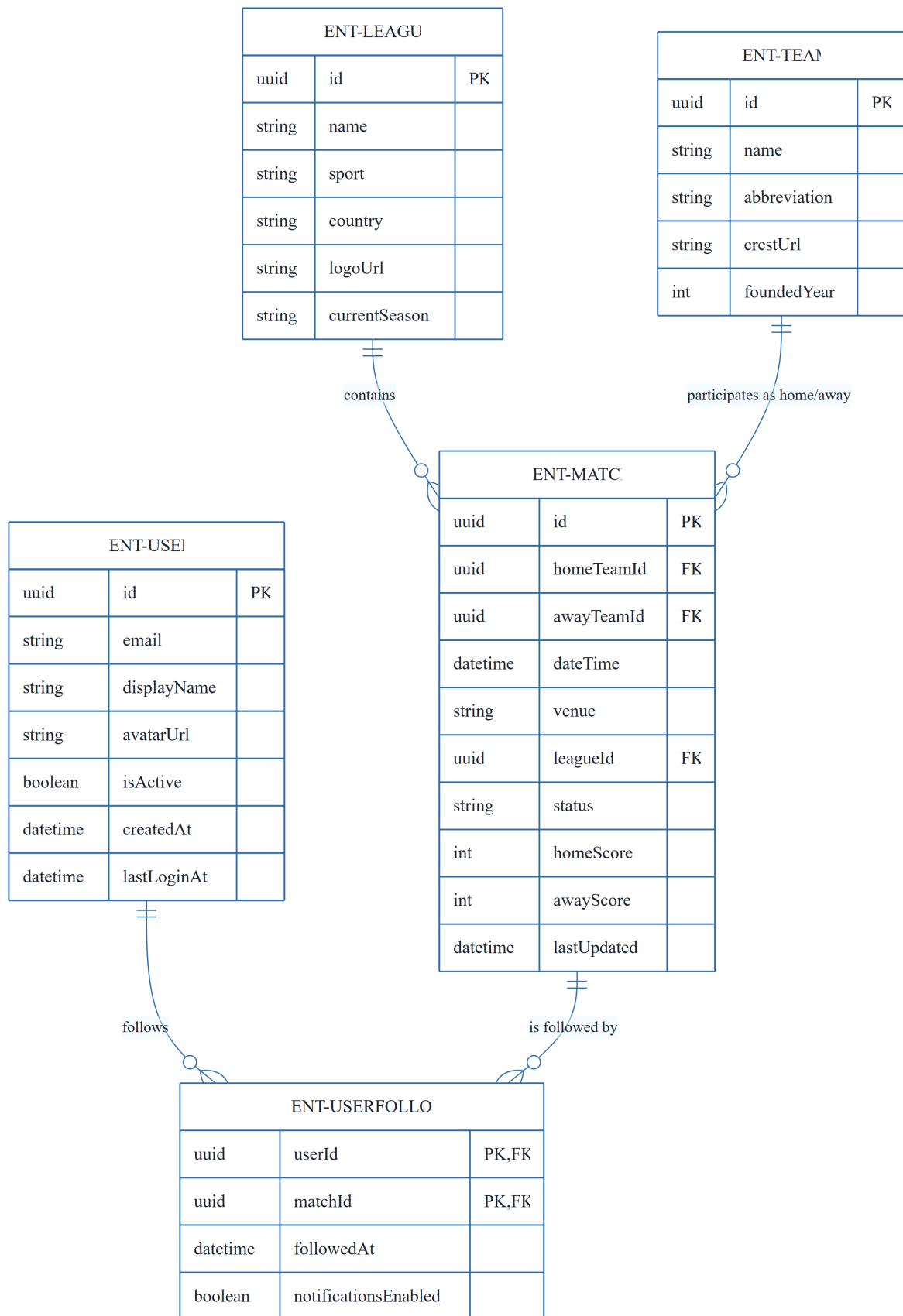
- **Match Integrity:** homeTeamId must not equal awayTeamId.
- **Temporal Logic:** dateTime must be in the future for matches with 'upcoming' status.
- **State-Based Validation:** homeScore and awayScore must be non-null when status is 'finished'.
- **Value Constraints:** homeScore and awayScore minimum value: 0.
- **Uniqueness:** Team names must be globally unique; User emails must be unique.
- **Relational Constraints:** UserFollow requires a composite unique constraint on (userId, matchId).
- **Field Limits:**
 - Team abbreviation: max 5 characters.
 - Venue: max 200 characters.
 - Team/League name: max 100 characters.

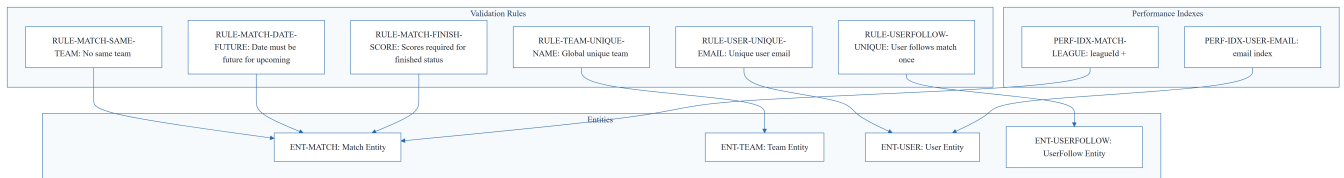
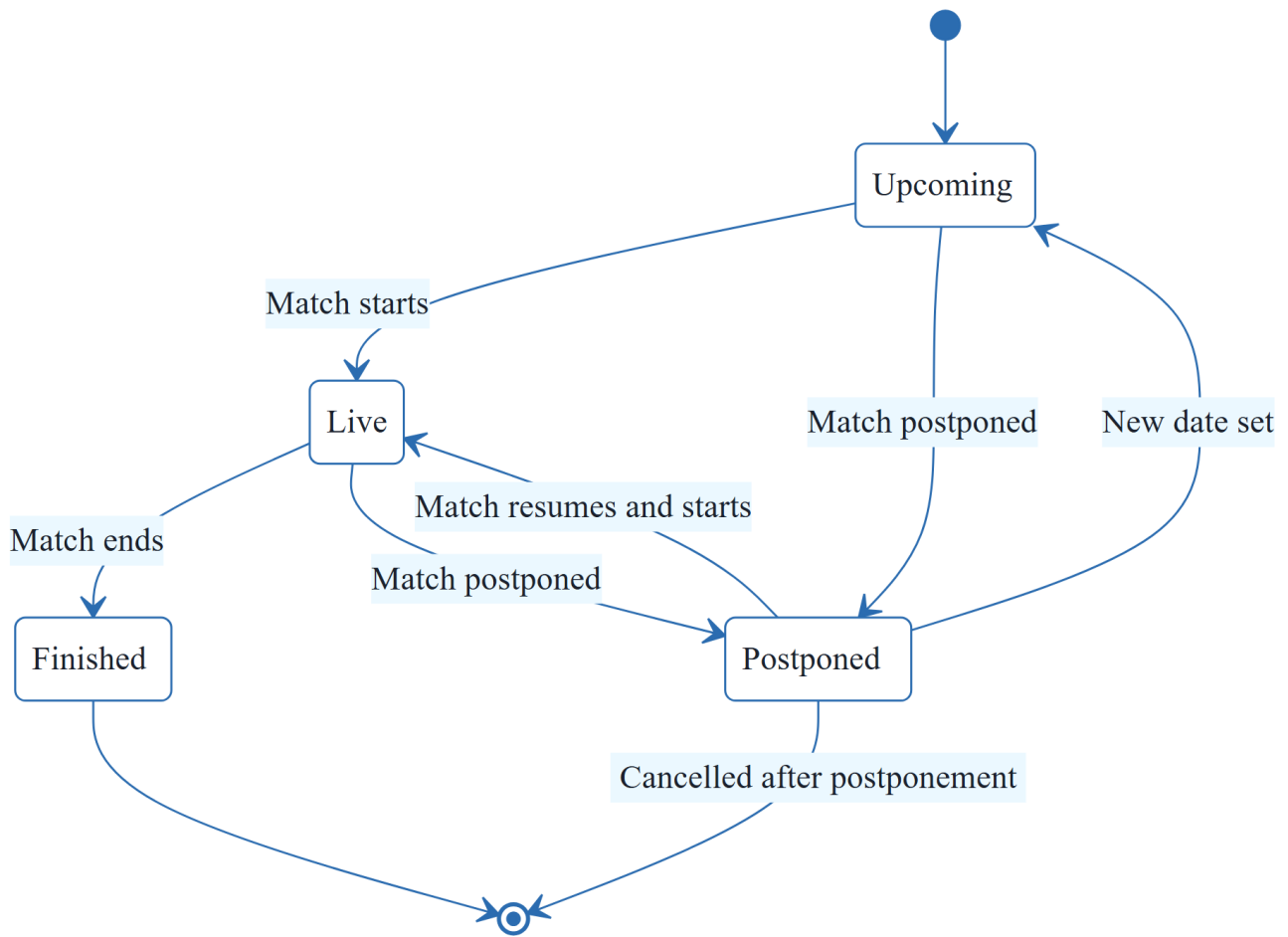
- User display name: max 50 characters.
- Team foundedYear: range 1800 to current year.

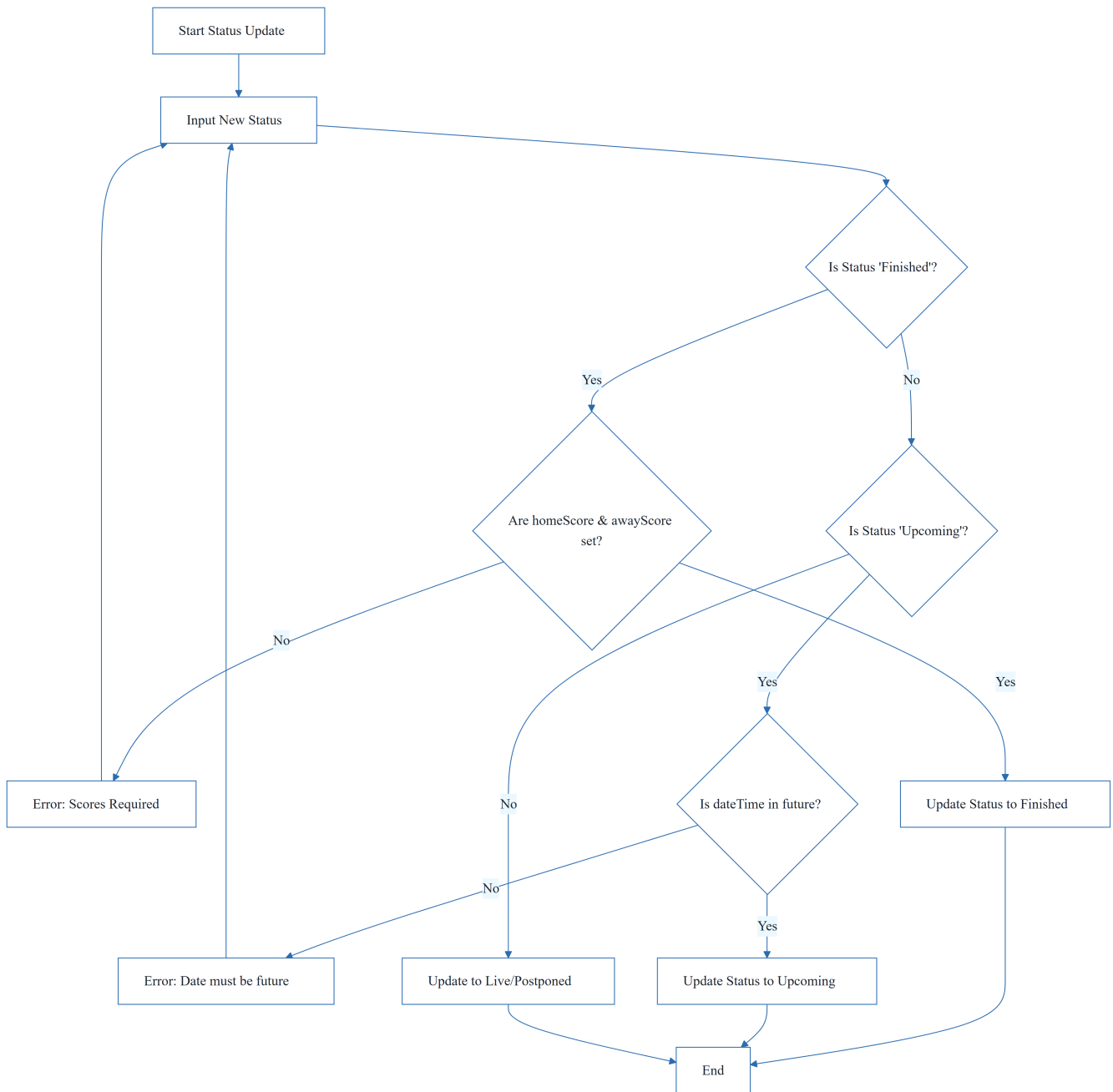
1.5 Critical Dependencies

- **Foreign Key Dependencies:** ENT-MATCH depends on ENT-LEAGUE and ENT-TEAM.
- **Relational Dependencies:** ENT-USERFOLLOW depends on ENT-USER and ENT-MATCH.
- **Performance Dependencies:**
 - Index on (leagueId, dateTime) for match filtering.
 - Index on (email) for User authentication.
 - Index on (userId) and (matchId) for UserFollow relational queries.

2. Architecture Workflows

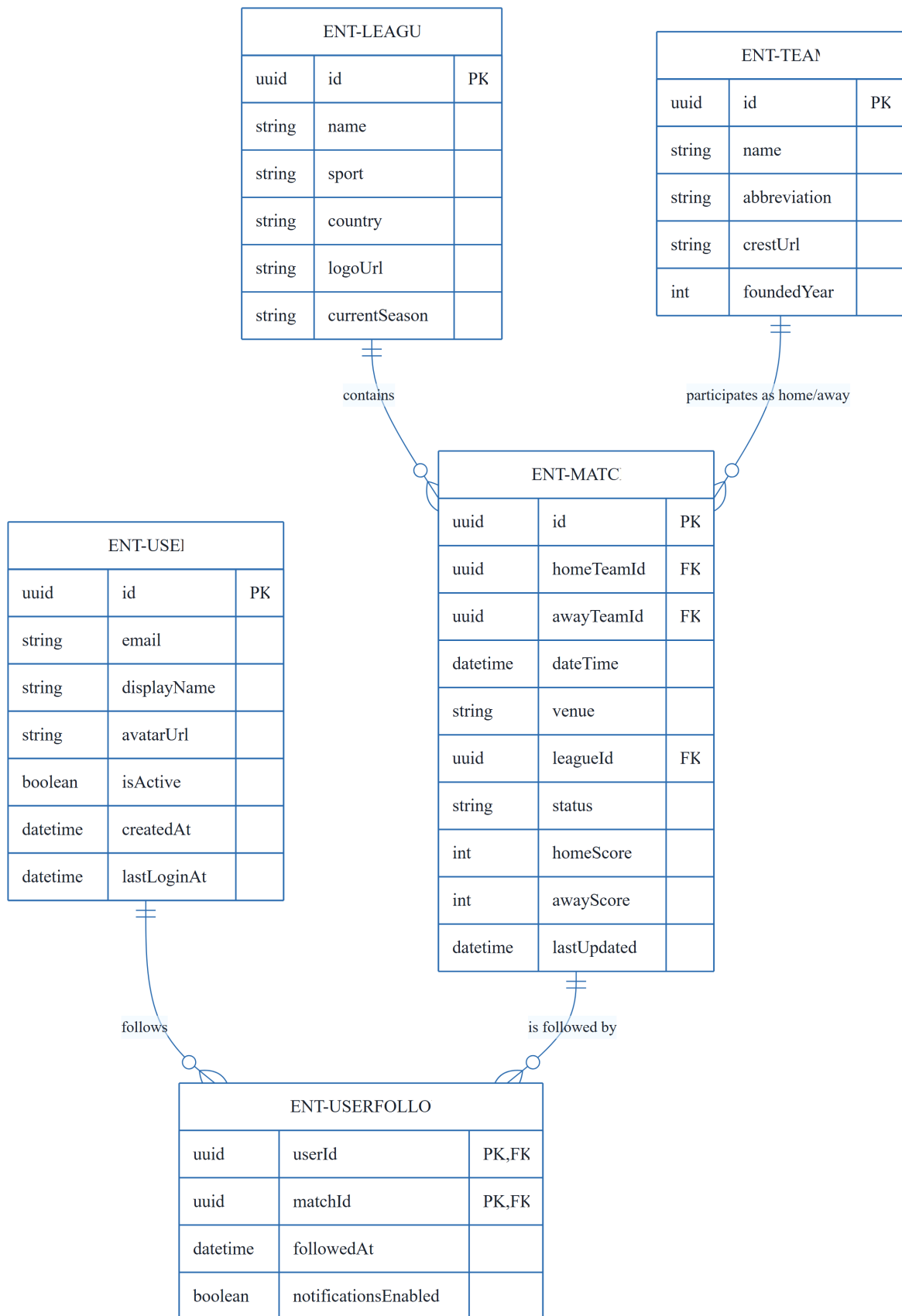




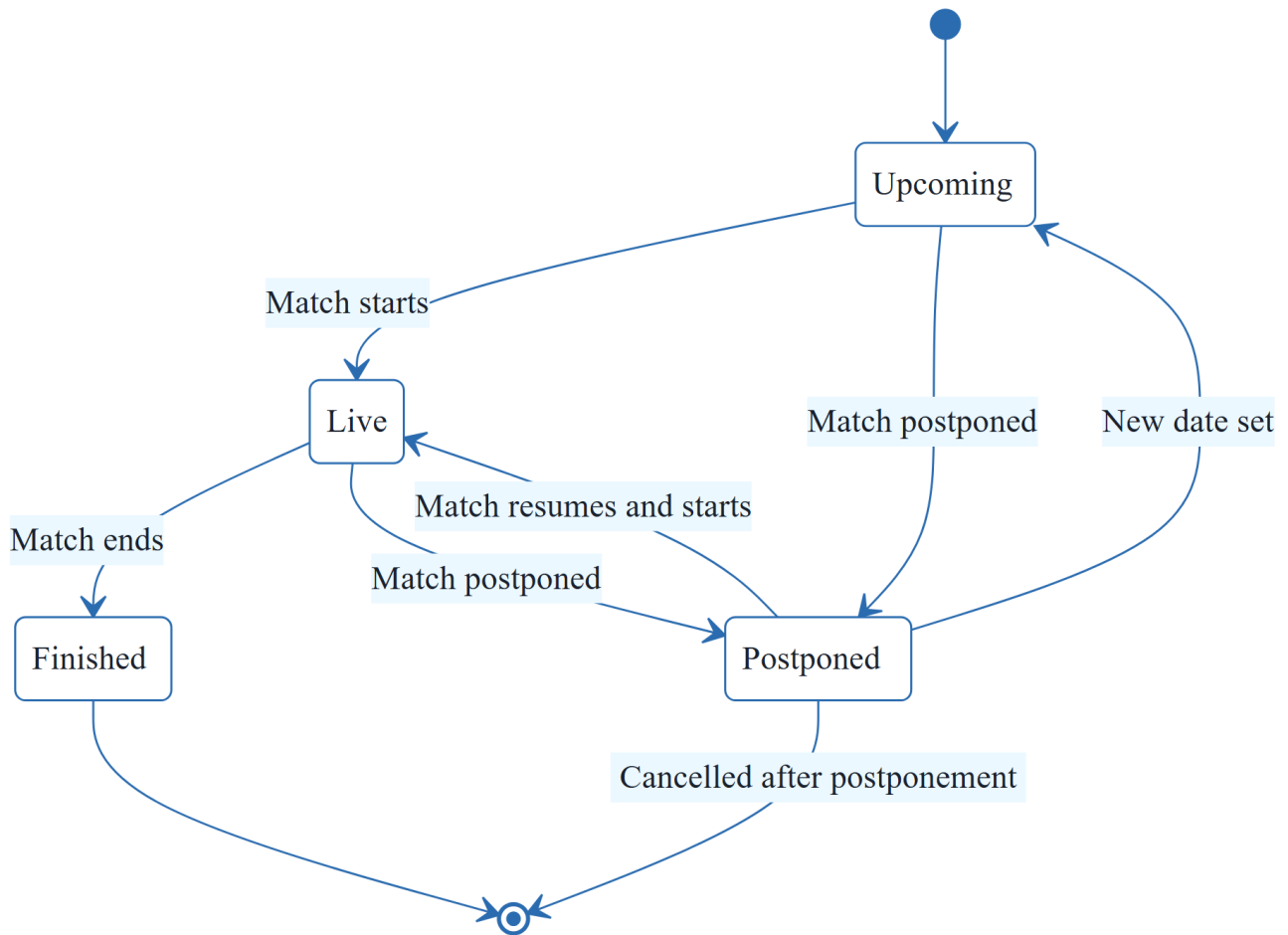


& Visual Diagrams

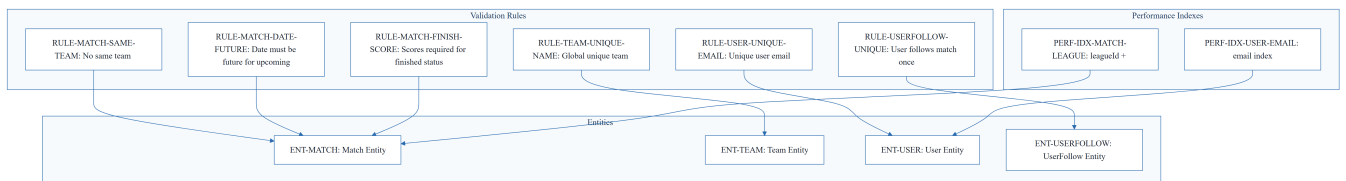
2.1 Entity Relationship Diagram



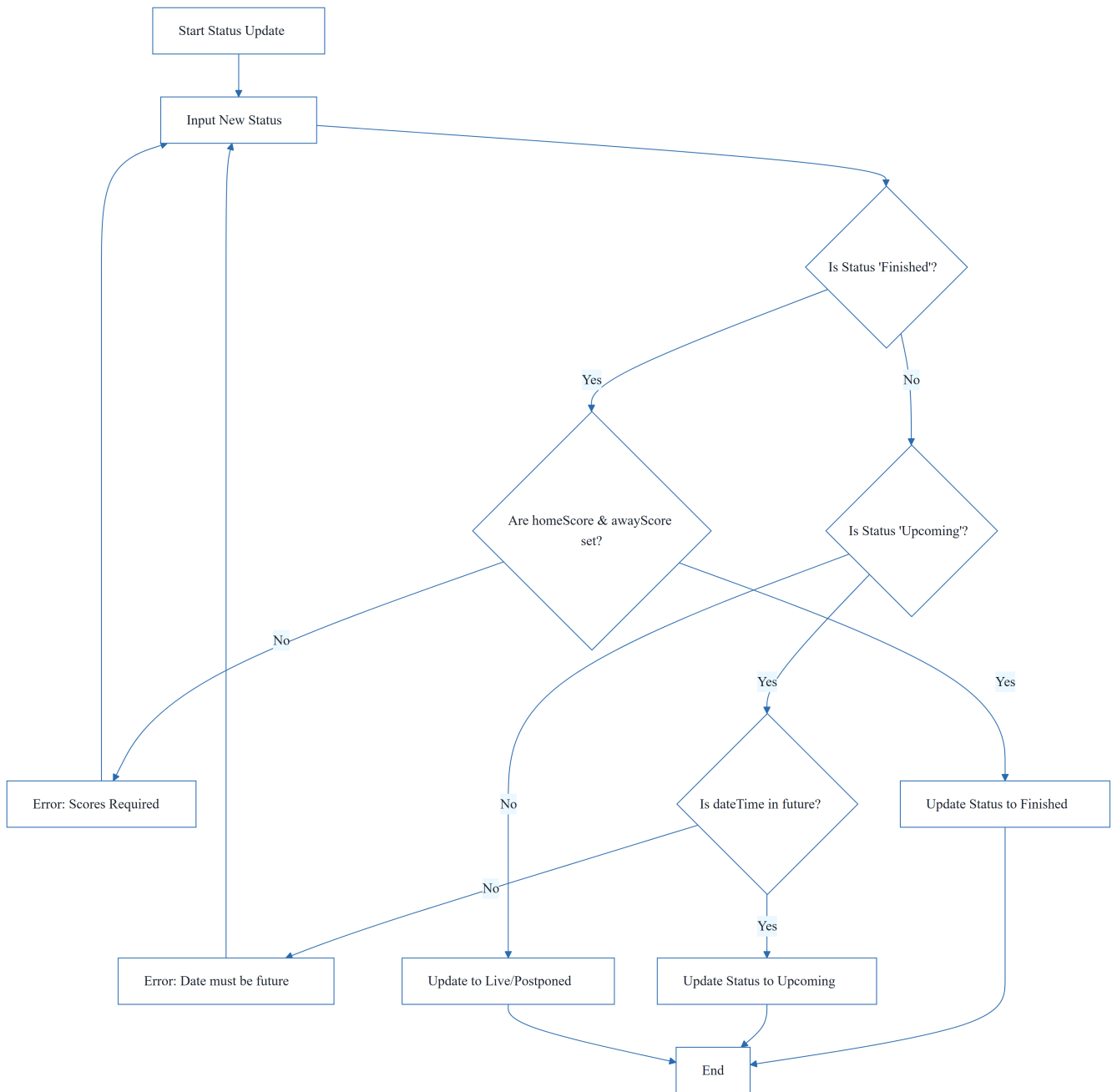
2.2 Match State Lifecycle



2.3 Data Validation & Performance Traceability



2.4 Match Status Update Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Type	Description	Source Section
ENT-MATCH	Entity	Football match entity containing details like teams, league, date, and score.	Match

ID	Type	Description	Source Section
ENT-TEAM	Entity	Football team entity including name, abbreviation, and crest.	Team
ENT-LEAGUE	Entity	League or tournament entity.	League
ENT-USER	Entity	Application user entity.	User
ENT-USERFOLLOW	Entity	Join entity representing the relationship between a User and a Match.	UserFollow
RULE-MATCH-SAME-TEAM	Constraint	A match cannot have the same team as both home and away.	Validation Rules
RULE-MATCH-DATE-FUTURE	Constraint	Match dateTime must be in the future for status 'upcoming'.	Validation Rules
RULE-MATCH-FINISH-SCORE	Constraint	When status changes to 'finished', both homeScore and awayScore must be set (non-null).	Validation Rules
RULE-SCORE-NON-NEGATIVE	Constraint	Scores cannot be negative.	Validation Rules
RULE-TEAM-UNIQUE-NAME	Constraint	Team name must be unique across all teams globally.	Validation Rules
RULE-USER-UNIQUE-EMAIL	Constraint	User email must be unique.	Validation Rules
RULE-USERFOLLOW-UNIQUE	Constraint	A user can only follow a match once (Composite unique constraint on userId, matchId).	Validation Rules
PERF-IDX-MATCH-LEAGUE	NFR	Index on (leagueId, dateTime) for filtering by league and date.	Indexes (for performance)
PERF-IDX-USER-EMAIL	NFR	Index on (email) for authentication.	Indexes (for performance)

3.2 Security Rules

- **Authentication:** User email must be unique (RULE-USER-UNIQUE-EMAIL).
- **Credential Integrity:** Password complexity requirements (min 8 chars) must be enforced.
- **Access Control:** User account status is managed via the `isActive` boolean flag.

3.3 Data Models

ENT-MATCH

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
homeTeamId	UUID	Reference to Team	Required, exists
awayTeamId	UUID	Reference to Team	Required, exists, != homeTeamId
dateTime	DateTime	Match date and time (UTC)	Required, future if 'upcoming'
venue	String (200)	Stadium or location name	Optional
leagueId	UUID	Reference to League	Required, exists
status	Enum	upcoming, live, finished, postponed, cancelled	Required, default: upcoming
homeScore	Integer	Goals scored by home team	Optional, min 0, default null
awayScore	Integer	Goals scored by away team	Optional, min 0, default null
lastUpdated	DateTime	Timestamp of last update	Auto-set on update

ENT-TEAM

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String (100)	Team name	Required, globally unique
abbreviation	String (5)	Team abbreviation	Optional, unique per league
crestUrl	URL	URL to team crest/logo	Optional, valid URL
foundedYear	Integer	Year the club was founded	Optional, 1800 <= year <= current

ENT-LEAGUE

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String (100)	League name	Required
sport	String	Sport type	Required, default: "football"
country	String (100)	Country where league is based	Optional
logoUrl	URL	URL to league logo	Optional, valid URL

Field	Type	Description	Validation
currentSeason	String	Current season identifier	Optional

ENT-USER

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
email	String (email)	User email address	Required, unique, valid format
displayName	String (50)	Display name for the user	Required
avatarUrl	URL	URL to user avatar/image	Optional, valid URL
isActive	Boolean	Whether the user account is active	Required, default: true
createdAt	DateTime	Account creation timestamp	Auto-set on create
lastLoginAt	DateTime	Last login timestamp	Updated on login

ENT-USERFOLLOW

Field	Type	Description	Validation
userId	UUID	Reference to User	Required, exists
matchId	UUID	Reference to Match	Required, exists
followedAt	DateTime	Timestamp of follow action	Required, auto-set on create
notificationsEnabled	Boolean	Notification preference	Optional, default: false

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Goals & Objectives	HIGH	The document describes 'How' the data is structured but not 'Why' or what the business objectives are.
Scope & Out-of-Scope	MEDIUM	Define what the football manager system will and will not handle (e.g., player stats, transfers).

Gap	Priority	Remediation Advice
Open Questions	LOW	Clarify the uncertainty mentioned in 'Team Validation' regarding global vs league uniqueness.

4.2 Remediation & Workflow

1. **Business Alignment:** Conduct a stakeholder interview to define the "Why" (Goals & Objectives).
2. **Scope Definition:** Explicitly list excluded features (e.g., "No player-level statistics") to prevent scope creep.
3. **Technical Decision:** Resolve the RULE-TEAM-UNIQUE-NAME ambiguity (Global vs League scope) and update the data model accordingly.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Constraints	TECHNICAL_STACK	RULE-USERFOLLOW-UNIQUE	The set of composite uniqueness requirements and logical limitations applied to data persistence to ensure integrity.
Cryptographic Hashing	TECHNICAL_STACK	RULE-USER-UNIQUE-EMAIL	The implied security mechanism for storing sensitive credentials to meet the mentioned complexity requirements.
DateTime	TECHNICAL_STACK	ENT-MATCH	A temporal data type used for scheduling events and tracking system audit stamps.
MUN	BUSINESS_DOMAIN	ENT-TEAM	A specific example of a short-form identifier assigned to a sports club.
Match Validation	BUSINESS_DOMAIN	Validation Rules	The logical checks ensuring teams are distinct, dates are future-dated for upcoming states, and scores are present upon completion.
Relationships	TECHNICAL_STACK	ENT-MATCH	The associative links between entities, such as the foreign key dependencies between competitions, teams, and schedules.

Term	Category	Context Anchor	Project Definition
Team Validation	BUSINESS_DOMAIN	RULE-TEAM-UNIQUE-NAM E	The requirement that club names must remain globally distinct within the database.
UTC	TECHNICAL_STACK	ENT-MATCH	The standardized time reference used to eliminate offset ambiguity for global event scheduling.
UUID	TECHNICAL_STACK	ENT-MATCH	A 128-bit universally unique identifier used as the primary key for all main entities.
User Validation	BUSINESS_DOMAIN	RULE-USER-UNIQUE-EMA IL	The verification process ensuring electronic mail addresses are unique and credentials meet minimum character lengths.
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	A join entity linking a person to a specific sporting event they wish to monitor.
UserFollow Validation	BUSINESS_DOMAIN	RULE-USERFOLLOW-UNI QUE	The logic preventing a single person from creating multiple subscriptions to the same sporting event.